# Sum of Subarray Minimums (medium)

## Problem Statement

Given an integer array `arr`, find the `sum` of the `minimum` elements of all possible `subarrays` of `arr`. Return the answer modulo $10^9 + 7$ as it may be very large.

### Examples

**Example 1**:

- **Input**: [2, 3, 1]
- **Expected Output**: 10
- **Justification**: The subarrays are [2], [3], [1], [2, 3], [3, 1], and [2, 3, 1]. Their minimums are **2**, **3**, **1**, **2**, **1**, and **1** respectively. Summing these minimums yields **10**.

**Example 2**:

- **Input**: [4, 5, 2]
- **Expected Output**: 19
- **Justification**: The subarrays are [4], [5], [2], [4, 5], [5, 2], and [4, 5, 2]. Their minimums are **4**, **5**, **2**, **4**, **2**, and **2** respectively. Summing these minimums gives us **19**.

**Example 3**:

- **Input**: [7, 2, 8, 4]
- **Expected Output**: 35
- **Justification**: The subarrays are [7], [2], [8], [4], [7, 2], [2, 8], [8, 4], [7, 2, 8], [2, 8, 4], and [7, 2, 8, 4]. Their minimums are **7**, **2**, **8**, **4**, **2**, **2**, **4**, **2**, **2**, and **2** respectively. Summing these minimums totals **35**.

## Try it yourself

Try solving this question here:

Java

```java
import java.util.Stack;

public class Solution {
    public int sumSubarrayMins(int[] arr) {
        // ToDO: Write Your Code Here.
        return 0; // Return the final sum.
    }
}

```

Test Case | Run Code Result | Stdout

[2, 3, 1]

Use Example Testcases | Run | Submit

← Previous
Solution: Delete Node In A BST

Next →
Solution: Sum Of Subarray Minimums

Mark as Completed